

A programmed solution to a problem

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Information about Mulligan Motors

- Mulligan Motors is an upcoming and ever-growing business and opened in year 2003 and is looking to expand even more.
- It was founded by the Mulligan family with Damien Mulligan as Managing Director and for the past 17 years has been selling high quality second-hand cars with the goal of providing the area with a unique service aimed at gaining customer loyalty.
- It's now one of the fastest developing/growing second hand auto dealership within the Newry and Belfast area with 5 thousand+ cars sold thus far
- The business ensures customers remain happy by generously providing full warrantees for each car they sell which are first quality approved in their workshop beforehand.

Key goals of my solution

- A database that provides an intuitive interface with a set of navigation tools that allow quick and easy access throughout the system.
- The ability to append to, delete from and update individual records held across a variety of tables.
- Each record inputted will be reasonably validated.
- Have a fully functioning relationship system that enable the database to be able to access and retrieve data from within itself.
- A secure database that allows the customers rights and follows the legislations and laws in place in the UK regarding the storage of data.

Primary Stakeholders

- Managers

They are accountable for overseeing stock, managing employees and making long and short-term decisions about the company. They should be able to set levels of access and create accounts for new employees.

- Frontline Salesman

These a the employees who interact with customers and the physical building on a daily basis. The database and the productivity increaseses should affect the frontline staff the most

- Operations staff

Staff who are involved in the business but work behind the scenes such as advertisers and those who purchase cars to sell. The ease of viewing what cars currently available and filling in forms for new cars in the new database should help these stakeholders massively.

Essential/Non-Essential Requirements:

Management

Essential

- A user-friendly and straightforward interface that enables the manager to effectively manage the database features and effortlessly navigate and use the database
- Secure access levels coupled with easily recognizable management features to control the dissemination of information.
- Database tools that facilitate easy and efficient retrieval of data stored within the database."
- A list of all employee's usernames and passwords.

Non-Essential

- For a simple guide to be included in the system to help the manager quickly adapt to the system when first using the database.

Questions for management:

1. Are there any additional requirements you deem essential to the database and believe should be added?
2. Are there any features or requirements of the system you deem unnecessary or wasteful of resources?
3. Are there any features or requirements you deem unessential to the database system but would increase the standard of delivery?
4. Are there any aesthetic features you feel should the database should have?

Essential /Non-Essential Requirements:

Frontline-Salesmen

Essential

- A simplistic intuitive interface that allows the staff to utilise all features of the database within a timely manner to maxamise sales.
- Quick access to data held within the database with a simple login page and identifiable buttons and headers.
- A page displaying sales information about the employee since a selected time frame and the total amount of commission made.
- Ability to book a non-conflicting test drive for a customer.
- Ability to quickly search customer records using non-key attributes such as First name & Last name.

Non-Essential

- To allow a salesperson to see their ranking amongst other employees in terms of sales revenue made.
- For a simple guide to be included in the system to help train frontline salespeople to use the system when first using

Questions for Salesmen

1. Are there any additional requirements you deem essential to the database and believe should be added?
2. Are there any features or requirements of the system you deem unnecessary or wasteful of resources?
3. Are there any features or requirements you deem unessential to the database system but would increase the standard of delivery?

Essential /Non-Essential Requirements:

Operations staff

Essential

- A simplistic intuitive interface that allows the staff to utilise all features of the database within a timely manner to maxamise sales.
- Quick access to data held within the database with a simple login page and identifiable buttons and headers.
- A page showing the current inventory of cars available and how long they've been in stock for.
- Able to quickly and easily fill in information about new cars in stock.

Non-Essential

- For a simple guide to be included in the system to help train the operations staff to use the system.

Questions for Salesmen

1. Are there any additional requirements you deem essential to the database and believe should be added?
2. Are there any features or requirements of the system you deem unnecessary or wasteful of resources?
3. Are there any features or requirements you deem unessential to the database system but would increase the standard of delivery?

Constraints of the planned solution

- Firstly, the newness and excitement of a new system may distract and inhibit productivity in the short-term after the implementation of the project is complete.
- It will be up to the company to decide where, and how often they store backups of the digital system. This is an undecided variable in the budget.
- Whilst I aim to prevent this from happening, if the company accidentally leaks its own information, or deletes data without a backup, it will be the liability of the company, Mulligan Motors, to ensure this does not occur.
- It will likely take significant resources and time to transfer all the data from the paper based system to the digital medium

Constraints of the planned solution Continued

- If passwords are lost, there are no built in functions for an employee on his own to get his password and/or username back in writing.(Such as a Email verification) He would need to get the manager to give him his username back.
- The system will have no online/internet compatibility; it will remain offline. This may restrict its functionality.
- Certain forms such as inputting car information will still need to be done manually on a keyboard. It will be the employee's responsibility to prevent data integrity failure.

Programming Outline

- All the coding for the database will be done via python.
- Any GUI work will be completed using the TKinter module for python. It is well documented and tested due to its age (released in 1991) . This is a low risk method of developing a GUI as any major errors in the system will be relatively easy to troubleshoot for.
- The structure and relationships of the database will be crafted using SQLite3, which will add no cost to the budget as it is free and efficient storage wise.
- Passwords will be required to be a certain complexity to create an employee account.
- Levels of access will ensure only certain information will be available to certain staff e.g managers. Ideally, i hope to encrypt this information so those without access cannot see the information at all; even outside the program. However, I have never encrypted before and this may be too time consuming to do in the final solution.

Feedback

- I sent the questions separately for the manager, frontline staff, and operational staff through a google document form. I sent them a link after showing them my planned solutions and asked them to complete the questions to the best of their abilities.

Manger feedback

1.Are there any additional requirements you deem essential to the database and believe should be added?

N/A

2.Are there any features or requirements of the system you deem unnecessary or wasteful of resources?

N/A

3.Are there any features or requirements you deem unessential to the database system but would increase the standard of delivery?

Our logo is blue so please make the database aqua ultra marine blue themed.

Submitted 12/11/2024, 08:47

Operational staff feedback

1.Are there any additional requirements you deem essential to the database and believe should be added?

N/A

2.Are there any features or requirements of the system you deem unnecessary or wasteful of resources?

Backups will only slow down the system

3.Are there any features or requirements you deem unessential to the database system but would increase the standard of delivery?

The manager should be able to set bonuses when a certain target is reached

Submitted 12/11/2024, 08:54

Front-line salespeople feedback

1.Are there any additional requirements you deem essential to the database and believe should be added?

A system should be in place to automatically backup the data for us.

2.Are there any features or requirements of the system you deem unnecessary or wasteful of resources?

All the employees can be trusted, there is no need to encrypt data. Levels of access is enough security.

3.Are there any features or requirements you deem unessential to the database system but would increase the standard of delivery?

A training section will only take up space, a seperate video that goes through the program would be better

Submitted 12/11/2024, 10:02

Response to feedback

1." A system should be in place to automatically backup data for us"

No, whilst this is a useful feature that would decrease the maintenance involved with the system, it is unfeasible in my current time frame

2. "Backups will only slow down the system"

I do not concur, the option to implement backups will have little effect on the system especially if done at an off-peak time such as just after close. This is as very little staff will be using the system when the backup occurs.

3. "All the employees can be trusted, there is no need to encrypt data. Levels of access is enough security."

Whilst I have no doubt that the employees wouldn't go out of their way to have unauthorised access to data, encryption will stop malicious actors who target Mulligan Motors to leak their data.

4. "Our logo is blue so please make the database aqua ultra marine blue themed."

I agree, I will aim to make the database look aesthetically pleasing, matching the colour theme of forms and reports to match Mulligan Motors logo is a good idea

5. "The manager should be able to set bonuses when a certain target is reached"

Whilst this is a good idea, is is non-essential and its addition to the system is dependant on how much spare time I have at the end of the development cycle.

6. "A training section will only take up space, a seperate video that goes through the program would be better"

This is a good suggestion, not only will this reduce bloat and storage space in the system, but would likely be easier to implement rather than coding a tutorial in python. I will likely implement a video to train the staff instead of a built in training program.